Toward Fortran Level AMICA in Julia

Block Based Learning and GPU Acceleration

Valentin Morlock, 8.10.2025

Agenda

- **Independent Component Analysis**
- **ICA for EEG data**
- **AMICA & AMICA.jl**
- **Problem**
- **Research Question**
- **Approach**

About Me

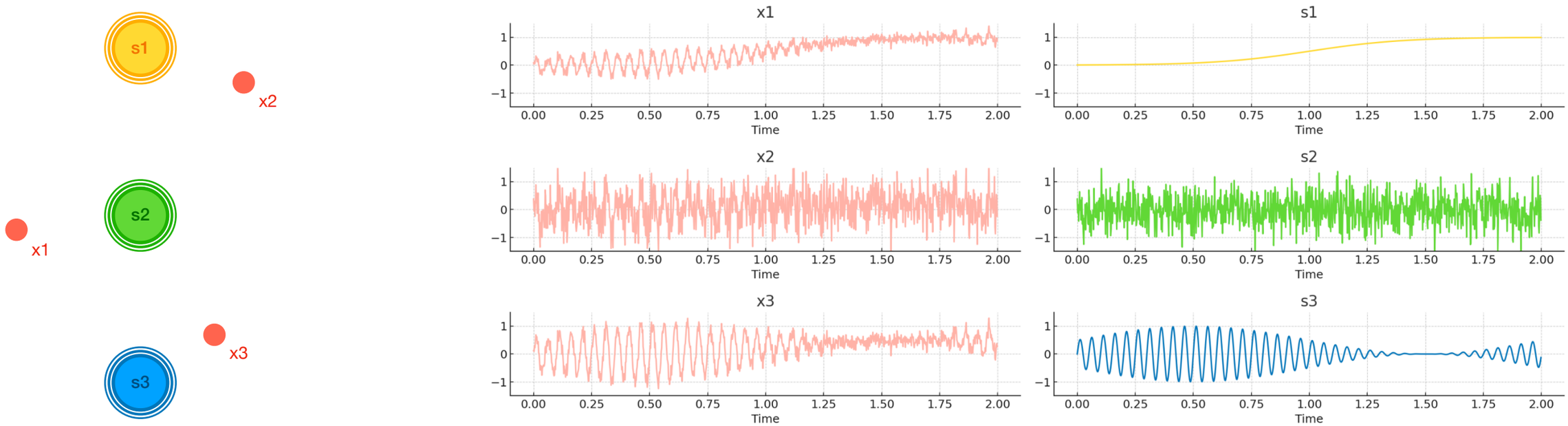
- B.Sc. in Business Information Systems in Augsburg
- 5 years of work experience in Software Engineering
- M.Sc. in Software Engineering

Independent Component Analysis

- ICA separates mixed signals into original sources (e.g. cocktail party problem).

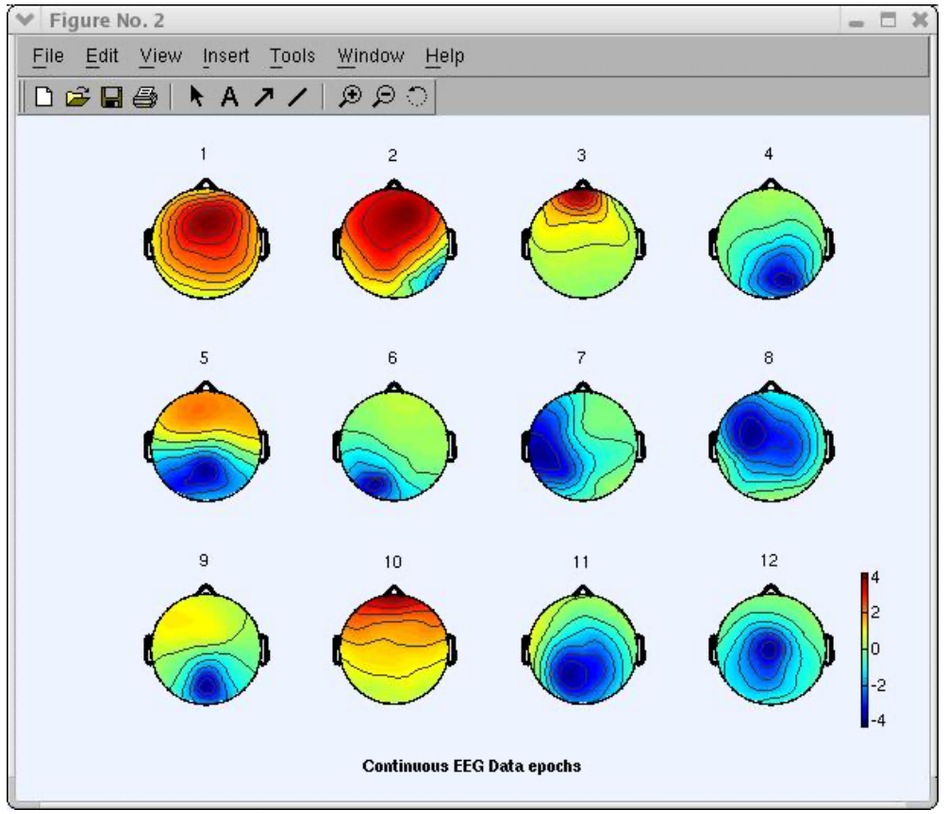
- Linear transformation from source signals s to mixtures x
  $x = A * s$ → estimate $W = A^{-1}$ to reverse mixing $s = x * W$

- Approached e.g. by maximizing entropy or maximizing likelihood



ICA for EEG data

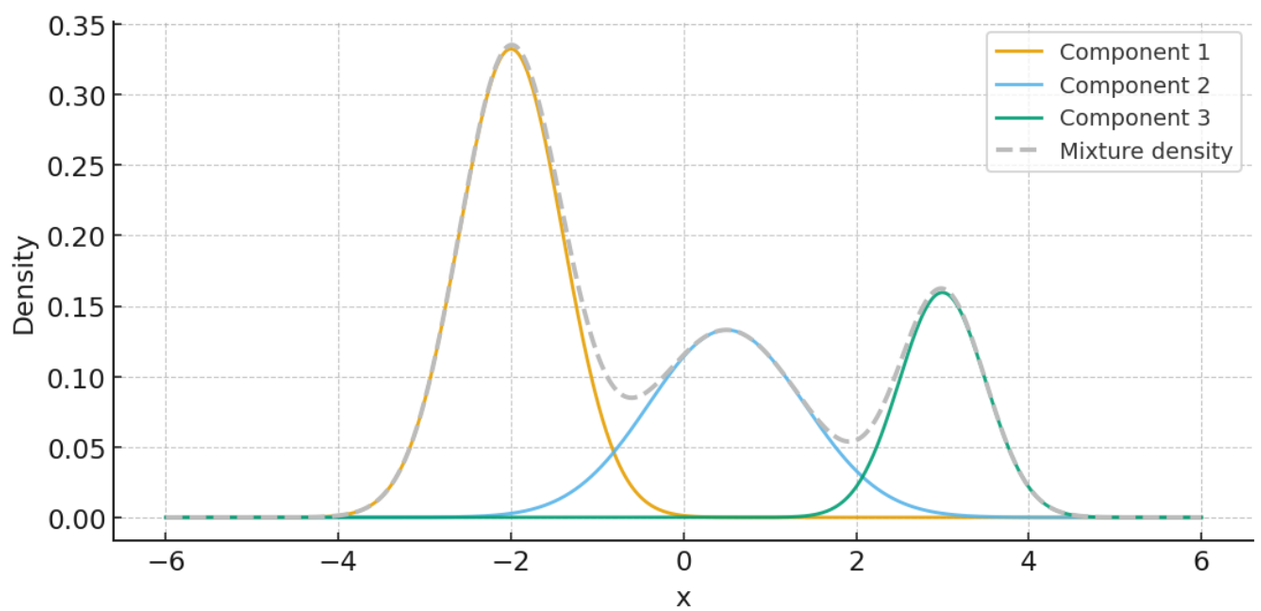
- EEG signals are mixtures of many overlapping brain and non-brain sources
- ICA can separate these mixtures into independent components
- Used to remove Artifacts (e.g. eye blinks, heartbeats) or separate brain sources
- AMICA has shown to be most successful in separating EEG sources



https://eeglab.org/tutorials/06_RejectArtifacts/RunICA.html

AMICA

- Maximum-likelihood based ICA
- Learns source densities as mixtures of Gaussian Scale Mixtures
- Multiple models for non-stationarity



Problem

- Three implementations with distinct problems

	Maintainability	Convergence	Performance
Fortran	--	++	++
Matlab	+	-	--
Julia	++	-	--

Research Question

- Can an optimized Julia AMICA match Fortran in convergence and runtime efficiency?
- Which improvements (e.g. blockwise learning, GPU acceleration) show the strongest contribution?

Approach

- Add a comprehensive test suite
- Investigate and add blockwise learning
- Add GPU support
- Extend documentation and tutorials